

# NORA Voice Agent — Test Report

Automated regression suite, simulated call tests, live API checks and a manual live call. Generated from the recorded pytest run; every status below is the actual recorded outcome.

| Item                     | Detail                                                                                                                                                  |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| System under test        | NORA — outbound payment/service reminder voice agent (FastAPI backend)                                                                                  |
| Modules covered          | app/main.py (turn loop), app/services/policy.py, app/services/intent.py, app/services/llm.py, app/services/tts.py, app/utis/language.py, app/session.py |
| Test date                | 5 October 2026                                                                                                                                          |
| Environment              | Windows 10, Python 3.10, FastAPI 0.135.1, pytest 9.0.2                                                                                                  |
| External services        | Groq qwen/qwen3.8-27b (reply + intent), Deepgram nova-2 (STT), Rime arcana (TTS)                                                                        |
| Automated cases executed | 409 (409 passed, 0 failed); suite runtime 41.5 s                                                                                                        |
| Build                    | working tree (uncommitted), audio cache manifest: 125 clips                                                                                             |

## 1. Test approach

Testing was carried out at four levels so that call behaviour could be verified exhaustively without placing real phone calls for every scenario.

| Level              | What it covers                                                                                                                              | How external services are handled                                                                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1 — Unit          | Intent recognition rules, callback-time parsing, prompt construction, session state transitions.                                            | No external calls; pure functions.                                                                                                                               |
| L2 — Component     | The policy engine: one decision per conversational turn, including every compliance, escalation and conversation-control path.              | No external calls; the engine is deterministic by design.                                                                                                        |
| L3 — Integration   | The real WebSocket turn loop in main.py end to end: audio in → endpointing → filters → classification → policy → spoken reply → call close. | Speech-to-text, text-to-speech, the language model and the voice-activity detector are replaced with controllable fakes; all application logic is the real code. |
| L4 — Live / manual | A real browser call through the running server, plus live API checks against Groq.                                                          | Real Deepgram, Groq and Rime services.                                                                                                                           |

## 2. Results summary

| Test area                              | Suite                | Cases | Passed | Failed | Result |
|----------------------------------------|----------------------|-------|--------|--------|--------|
| Intent recognition (L1)                | intent_heuristics.py | 48    | 48     | 0      | PASS   |
| Call-flow policy engine (L2)           | policy.py            | 86    | 86     | 0      | PASS   |
| System prompt construction (L1)        | prompt.py            | 247   | 247    | 0      | PASS   |
| Session state & speculation guard (L1) | turn_flow.py         | 17    | 17     | 0      | PASS   |
| Language-model failure handling (L1)   | llm_failures.py      | 3     | 3      | 0      | PASS   |
| Full call simulation (L3)              | ws_simulation.py     | 8     | 8      | 0      | PASS   |
| TOTAL                                  |                      | 409   | 409    | 0      | PASS   |

The suite runs offline in about 40 seconds and is repeatable: it was executed four consecutive times with identical results while stabilising a timing-dependent case.

## 3. Main functional test cases

These cover the normal path of a reminder call from greeting to close.

| ID      | Test condition                                   | Input / steps         | Expected result                                                                       | Verified by                                                                        | Status |
|---------|--------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------|
| TC-F-01 | Customer confirms identity after the greeting    | "yes speaking"        | Identity marked confirmed; call stage advances to REMIND; reply delivers the reminder | test_confirm_then_already_paid, test_confirmation_after_greeting_verifies_identity | PASS   |
| TC-F-02 | Reminder delivered then customer says it is paid | "I already paid that" | Call ends politely; outcome recorded as PAID; no language-model call needed           | test_confirm_then_already_paid, test_already_paid_closes                           | PASS   |

| ID      | Test condition                                           | Input / steps                                      | Expected result                                                                   | Verified by                                                          | Status |
|---------|----------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|--------|
| TC-F-03 | Customer asks what the call is about                     | "Can you tell me about the payment?"               | Classified ASK_MORE_INFO; handled by the language model within the stage goal     | test_heuristic_intent                                                | PASS   |
| TC-F-04 | Customer wants a different time                          | "I need to reschedule" then "next tuesday morning" | Time captured and read back; call closes with CALLBACK_SCHEDULED                  | test_reschedule_asks_for_time_then_confirms                          | PASS   |
| TC-F-05 | Customer is busy and names a time in the same breath     | "I'm busy, call me in two hours"                   | Callback confirmed immediately without asking again                               | test_busy_with_time_schedules_callback, test_busy_then_callback_time | PASS   |
| TC-F-06 | Customer wants the details in writing                    | EMAIL_REQUEST intent with an address on file       | Address read back for confirmation before the call closes                         | test_email_confirmation_instruction                                  | PASS   |
| TC-F-07 | Account details stay private until identity is confirmed | Prompt built at stage VERIFY                       | Prompt instructs that details are shared only after confirmation                  | test_account_details_are_gated_on_identity                           | PASS   |
| TC-F-08 | Every call ends with a spoken goodbye                    | All 19 call-ending lines                           | Each ends with a closing phrase and never with a question                         | test_every_call_ending_line_says_goodbye                             | PASS   |
| TC-F-09 | Normal conversational turn reaches the language model    | "what's this about?" after identity confirmed      | Policy returns LLM; no fixed line substituted                                     | test_normal_turn_goes_to_llm                                         | PASS   |
| TC-F-10 | Stage progresses only on real events                     | Slots set one at a time                            | INTRO → VERIFY → REMIND → RESOLVE → CLOSE derived from call state, not turn count | test_session_stage_progression_and_latency_marks                     | PASS   |

## 4. Edge-case and compliance matrix

Each row is an automated case. Compliance and safety situations end the call with a fixed line and never reach the language model, so their wording cannot drift.

### Compliance — call must stop

| ID      | Scenario / input                                                               | Expected result                                                     | Verified by                                                                         | Status |
|---------|--------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|
| TC-E-01 | "please stop calling me"                                                       | Call ends, number recorded for suppression, disposition DO_NOT_CALL | test_do_not_call_is_recorded_for_suppression, test_compliance_stops_end_without_llm | PASS   |
| TC-E-02 | "you have the wrong number"                                                    | Apology, record marked, call ends                                   | test_compliance_stops_end_with_out_llm                                              | PASS   |
| TC-E-03 | "talk to my lawyer"                                                            | Call ends, ATTORNEY_REPRESENTED recorded                            | test_compliance_stops_end_with_out_llm                                              | PASS   |
| TC-E-04 | "I filed for bankruptcy"                                                       | Call ends, BANKRUPTCY recorded                                      | test_compliance_stops_end_with_out_llm                                              | PASS   |
| TC-E-05 | "she passed away last month"                                                   | Condolence, calls stopped, DECEASED recorded                        | test_compliance_stops_end_with_out_llm                                              | PASS   |
| TC-E-06 | "sorry my english is not good"                                                 | Call ends politely, LANGUAGE_BARRIER recorded                       | test_compliance_stops_end_with_out_llm                                              | PASS   |
| TC-E-07 | "don't call me at work" / "just text me instead"                               | Preference stored, not treated as a refusal                         | test_partial_opt_out_records_preference                                             | PASS   |
| TC-E-08 | Refusal offer answered with "just stop them"                                   | Treated as a full opt-out; DO_NOT_CALL                              | test_not_interested_then_stop_is_do_not_call                                        | PASS   |
| TC-E-09 | Hard-stop phrase hidden in a sentence: "my account is active but stop calling" | Compliance rule wins over the answer shortcut                       | test_classify_intent_fast_path_precedence                                           | PASS   |

### Safety and abuse

| ID      | Scenario / input               | Expected result                                                | Verified by                           | Status |
|---------|--------------------------------|----------------------------------------------------------------|---------------------------------------|--------|
| TC-E-10 | "I want to hurt myself"        | Crisis line (911 / 988) read out, escalation logged, call ends | test_distress_ends_with_crisis_line   | PASS   |
| TC-E-11 | Profanity once                 | One calm de-escalation offer                                   | test_abuse_deescalates_once_then_ends | PASS   |
| TC-E-12 | Profanity twice                | Call ends, disposition ENDED_ABUSE                             | test_abuse_deescalates_once_then_ends | PASS   |
| TC-E-13 | Abuse then "just end the call" | Immediate polite close                                         | test_abuse_then_end_request_closes    | PASS   |

### Escalation (callback, no live transfer)

| ID      | Scenario / input                                                             | Expected result                                       | Verified by                                                     | Status |
|---------|------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------|--------|
| TC-E-14 | Asks for a human / disputes the amount / hardship / payment plan / complaint | Callback from the team offered and reason recorded    | test_escalations_offer_callback                                 | PASS   |
| TC-E-15 | "can I pay over the phone"                                                   | Payment details refused on the call; callback offered | test_pay_now_never_takes_payment_and_caps_callback_asks         | PASS   |
| TC-E-16 | Callback offered, caller declines                                            | Call ends politely; ESCALATION_DECLINED               | test_escalation_declined                                        | PASS   |
| TC-E-17 | Caller never gives a time (asked twice)                                      | Stops asking; promises a business-hours callback      | test_pay_now_never_takes_payment_and_caps_callback_asks         | PASS   |
| TC-E-18 | Hardship plus a time in one sentence                                         | Callback confirmed in the same turn                   | test_escalation_with_time_in_same_sentence_confirms_immediately | PASS   |

### Right party / identity

| ID      | Scenario / input   | Expected result                                                     | Verified by                                       | Status |
|---------|--------------------|---------------------------------------------------------------------|---------------------------------------------------|--------|
| TC-E-19 | "this is his wife" | No account details disclosed; asks when to reach the account holder | test_third_party_gets_callback_without_disclosure | PASS   |

| ID      | Scenario / input                       | Expected result                                | Verified by                                                 | Status |
|---------|----------------------------------------|------------------------------------------------|-------------------------------------------------------------|--------|
| TC-E-20 | Third party answers "I don't know"     | Polite close instead of repeating the question | test_third_party_who_does_not_know_gets_polite_close        | PASS   |
| TC-E-21 | "yes" after "Is this a good time?"     | NOT treated as identity confirmation           | test_good_time_question_is_not_identity_confirmation        | PASS   |
| TC-E-22 | "okay" alone after the greeting        | NOT treated as identity confirmation           | test_okay_alone_does_not_confirm_identity                   | PASS   |
| TC-E-23 | "yes, who's calling?"                  | Identity confirmed and screening handled       | test_yes_with_screening_question_still_confirms_identity    | PASS   |
| TC-E-24 | Reply that never mentions the reminder | Reminder NOT marked delivered                  | test_reminder_not_marked_delivered_without_reminder_content | PASS   |

## Conversation control

| ID      | Scenario / input                              | Expected result                                                     | Verified by                                                                      | Status |
|---------|-----------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|
| TC-E-25 | "hold on" / "one sec"                         | NORA waits; silence timer suspended                                 | test_hold_checkin_and_timeout                                                    | PASS   |
| TC-E-26 | Hold lasting 30 s / 90 s                      | Check-in at 30 s, polite close at 90 s                              | test_hold_checkin_and_timeout                                                    | PASS   |
| TC-E-27 | Caller returns from hold                      | Hold cleared, conversation resumes                                  | test_any_reply_ends_hold                                                         | PASS   |
| TC-E-28 | "what?" / "say that again"                    | Last reply replayed without calling the language model              | test_repeat_replays_last_reply_then_offers_callback                              | PASS   |
| TC-E-29 | Third repeat request                          | Line quality assumed poor; callback offered                         | test_repeat_replays_last_reply_then_offers_callback                              | PASS   |
| TC-E-30 | "slow down"                                   | Speech speed reduced for the rest of the call and the line repeated | test_slow_down_lowers_speed_and_repeats                                          | PASS   |
| TC-E-31 | "are you a robot?"                            | Honest disclosure that it is an automated assistant                 | test_ai_question_is_answered_honestly, test_prompt_requires_honest_ai_disclosure | PASS   |
| TC-E-32 | Asked a second time, then declines a callback | Call continues instead of ending                                    | test_declining_callback_after_ai_disclosure_continues_the_call                   | PASS   |
| TC-E-33 | "bye"                                         | Call ends gracefully; not counted as a refusal                      | test_goodbye_ends, test_goodbye_is_not_a_refusal                                 | PASS   |
| TC-E-34 | Same question asked twice in a row            | Loop detected; callback offered instead of a third ask              | test_loop_breaker_after_same_question_twice                                      | PASS   |

## Follow-up accuracy while awaiting a callback time

| ID      | Scenario / input                        | Expected result                                 | Verified by                                          | Status |
|---------|-----------------------------------------|-------------------------------------------------|------------------------------------------------------|--------|
| TC-E-35 | "no, not tomorrow, friday works"        | Books Friday, not tomorrow                      | test_extract_callback_time_edge_cases                | PASS   |
| TC-E-36 | "good morning" / "at one point we paid" | Not mistaken for a callback time                | test_extract_callback_time_edge_cases                | PASS   |
| TC-E-37 | "anytime is fine"                       | Closes with a business-hours promise            | test_anytime_closes_with_business_hours_promise      | PASS   |
| TC-E-38 | "actually I already paid on tuesday"    | Outcome PAID wins over booking Tuesday          | test_new_outcome_while_waiting_for_time_wins         | PASS   |
| TC-E-39 | "what?" while a time was being asked    | Question stays open; a later time still books   | test_repeat_keeps_the_callback_question_open         | PASS   |
| TC-E-40 | "hold on" while a time was being asked  | Question stays open after the hold              | test_hold_keeps_the_callback_question_open           | PASS   |
| TC-E-41 | A new question instead of a time        | Pending ask dropped; question answered normally | test_new_question_while_pending_callback_goes_to_llm | PASS   |
| TC-E-42 | Still busy, no time given               | Closes politely rather than asking again        | test_still_busy_without_time_closes                  | PASS   |

## Soft refusals

| ID      | Scenario / input           | Expected result                               | Verified by                         | Status |
|---------|----------------------------|-----------------------------------------------|-------------------------------------|--------|
| TC-E-43 | Not interested, first time | One alternative offered (callback or opt-out) | test_busy_gets_one_offer_then_close | PASS   |

| ID      | Scenario / input              | Expected result                     | Verified by                                                                    | Status |
|---------|-------------------------------|-------------------------------------|--------------------------------------------------------------------------------|--------|
| TC-E-44 | Refused twice                 | Call closes; no further pushing     | test_busy_gets_one_offer_then_close, test_not_interested_then_no_closes_softly | PASS   |
| TC-E-45 | Bare "no" as an answer        | Treated as an answer, not a refusal | test_bare_no_is_not_a_refusal                                                  | PASS   |
| TC-E-46 | A number or date as an answer | Treated as an answer, not a refusal | test_numeric_answer_is_unclear_not_refusal                                     | PASS   |

## Telephony and system failures

| ID      | Scenario / input                                    | Expected result                                            | Verified by                                                                                       | Status |
|---------|-----------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|
| TC-E-47 | Voicemail greeting detected                         | Short call-back message left, then hang up; VOICEMAIL_LEFT | test_voicemail_leaves_message, test_classify_telephony_prompt                                     | PASS   |
| TC-E-48 | IVR menu detected                                   | Call ends, IVR_REACHED recorded                            | test_classify_telephony_prompt                                                                    | PASS   |
| TC-E-49 | Language model fails once                           | "Sorry, I missed that. Could you say it one more time?"    | test_first_failure_asks_caller_to_repeat                                                          | PASS   |
| TC-E-50 | Language model fails twice in a row                 | Call ends with a technical-trouble goodbye                 | test_second_failure_stays_quiet_for_main_to_end_call, test_repeated_llm_failure_ends_with_goodbye | PASS   |
| TC-E-51 | Language model fails during the wrap-up goodbye     | A goodbye is still spoken                                  | test_wrap_up_failure_still_says_goodbye                                                           | PASS   |
| TC-E-52 | Live speech synthesis returns no audio on a goodbye | Pre-recorded goodbye clip plays instead                    | test_goodbye_falls_back_to_cached_audio_when_tts_is_silent                                        | PASS   |
| TC-E-53 | Short but meaningful reply (2 words)                | Not discarded as a fragment                                | test_short_control_phrase_is_not_dropped_and_hold_resumes                                         | PASS   |

## 5. Intent recognition data set

Each utterance below was run through the recognition rules and checked against the expected intent.

| Utterance                           | Expected intent   | Status | Utterance                       | Expected intent | Status |
|-------------------------------------|-------------------|--------|---------------------------------|-----------------|--------|
| "I want to kill myself"             | DISTRESS          | PASS   | "I already paid that last week" | ALREADY_PAID    | PASS   |
| "please stop calling me"            | DO_NOT_CALL       | PASS   | "it's been paid"                | ALREADY_PAID    | PASS   |
| "Thanks, but take me off your list" | DO_NOT_CALL       | PASS   | "I don't owe you anything"      | DISPUTE         | PASS   |
| "remove my number"                  | DO_NOT_CALL       | PASS   | "that's the wrong amount"       | DISPUTE         | PASS   |
| "don't call me at work"             | PARTIAL_OPT_OUT   | PASS   | "I can't afford it right now"   | HARDSHIP        | PASS   |
| "just text me instead"              | PARTIAL_OPT_OUT   | PASS   | "can I set up a payment plan"   | PAYMENT_PLAN    | PASS   |
| "you have the wrong number"         | WRONG_NUMBER      | PASS   | "can I pay over the phone"      | PAY_NOW         | PASS   |
| "nobody here by that name"          | WRONG_NUMBER      | PASS   | "I need to reschedule"          | RESCHEDULE      | PASS   |
| "talk to my lawyer"                 | ATTORNEY          | PASS   | "call me back tomorrow"         | CALL_BACK_LATER | PASS   |
| "I filed for bankruptcy"            | BANKRUPTCY        | PASS   | "are you a robot"               | ASK_IF_AI       | PASS   |
| "she passed away last month"        | DECEASED          | PASS   | "is this a scam"                | TRUST_CONCERN   | PASS   |
| "sorry my english is not good"      | LANGUAGE_BARRIER  | PASS   | "how did you get my number"     | ASK_SOURCE      | PASS   |
| "screw you"                         | ABUSE             | PASS   | "this is his wife"              | THIRD_PARTY     | PASS   |
| "let me talk to a real person"      | TRANSFER_TO_HUMAN | PASS   | "she's not home right now"      | THIRD_PARTY     | PASS   |
| "hold on one sec"                   | HOLD_REQUEST      | PASS   | "who is this?"                  | GATEKEEPER      | PASS   |
| "one second"                        | HOLD_REQUEST      | PASS   | "I'm too busy right now"        | TOO_BUSY        | PASS   |
| "What?"                             | REPEAT_REQUEST    | PASS   | "not interested"                | NOT_INTERESTED  | PASS   |
| "sorry can you repeat that"         | REPEAT_REQUEST    | PASS   | "no"                            | UNCLEAR         | PASS   |
| "slow down please"                  | SLOW_DOWN         | PASS   | "yes"                           | UNCLEAR         | PASS   |
| "okay bye"                          | GOODBYE           | PASS   | "I'm good, how are you?"        | UNCLEAR         | PASS   |

## 6. Callback time parsing

| Caller said                      | Time extracted       | Status |
|----------------------------------|----------------------|--------|
| "after 5 pm"                     | after 5 pm           | PASS   |
| "call me next tuesday morning"   | next tuesday morning | PASS   |
| "in two hours"                   | in two hours         | PASS   |
| "tomorrow at 3"                  | tomorrow at 3        | PASS   |
| "no idea"                        | (no time found)      | PASS   |
| "yes"                            | (no time found)      | PASS   |
| "no, not tomorrow, friday works" | friday               | PASS   |
| "not tomorrow"                   | (no time found)      | PASS   |
| "good morning"                   | (no time found)      | PASS   |
| "at one point we paid"           | (no time found)      | PASS   |
| "around five pm"                 | five pm              | PASS   |
| "after 5:30"                     | after 5:30           | PASS   |
| "anytime is fine"                | any time             | PASS   |
| "whenever"                       | any time             | PASS   |

## 7. Voicemail / IVR detection

| Audio transcribed at call start                                 | Classified as | Status |
|-----------------------------------------------------------------|---------------|--------|
| "Please leave a message after the tone"                         | VOICEMAIL     | PASS   |
| "The person you are calling is not available to take your call" | VOICEMAIL     | PASS   |
| "For billing press 1"                                           | IVR           | PASS   |
| "Hello?"                                                        | None          | PASS   |

## 8. Full call simulation (integration)

These drive the real WebSocket endpoint with synthetic audio frames; only the external services are faked. Each one exercises endpointing, transcript filtering, classification, the policy engine, speech output and call closure in the same code path used on a live call.

| ID      | Scenario                                      | Expected result                                                                                            | Verified by                                                | Status |
|---------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|--------|
| TC-I-01 | Identity confirmed, then already paid         | Reminder delivered at the right stage; disposition ALREADY_PAID; closing line spoken; server ends the call | test_confirm_then_already_paid                             | PASS   |
| TC-I-02 | Do-not-call on the first turn                 | No language-model call at all; suppression record written; call ends                                       | test_do_not_call_ends_immediately_and_records              | PASS   |
| TC-I-03 | Two-word hold request then the caller returns | "one sec" is not discarded; "Sure, take your time." spoken; hold cleared on return                         | test_short_control_phrase_is_not_dropped_and_hold_resumes  | PASS   |
| TC-I-04 | Busy, then a callback time on the next turn   | Time captured; callback request logged; disposition CALLBACK_SCHEDULED                                     | test_busy_then_callback_time                               | PASS   |
| TC-I-05 | Voicemail greeting at the start of the call   | Message left; disposition VOICEMAIL_LEFT                                                                   | test_voicemail_leaves_message                              | PASS   |
| TC-I-06 | Language model unavailable for two turns      | Retry line first, then a technical-trouble goodbye; disposition TECH_TROUBLE                               | test_repeated_llm_failure_ends_with_goodbye                | PASS   |
| TC-I-07 | Speech synthesis silent on the goodbye        | Cached goodbye clip used so the call never ends in silence                                                 | test_goodbye_falls_back_to_cached_audio_when_tts_is_silent | PASS   |
| TC-I-08 | All greeting variants ask who is speaking     | Each greeting asks for the person by name and stays under 20 words                                         | test_every_greeting_asks_who_isSpeaking                    | PASS   |

## 9. Non-functional testing

### 9.1 Response latency measured on a live call

Measured from the end of the caller's speech. Instrumentation is built into the agent and written to the logs.

| Stage                           | Turn 1  | Turn 2  | Turn 3  | Comment                                  |
|---------------------------------|---------|---------|---------|------------------------------------------|
| Intent decided                  | 502 ms  | 1 ms    | 280 ms  | 0–1 ms when a deterministic rule matches |
| First audio to caller (filler)  | 544 ms  | 27 ms   | —       | Cached audio masks the wait              |
| Language model first token      | 871 ms  | 335 ms  | 629 ms  | Groq qwen/qwen3.8-27b                    |
| First spoken words of the reply | 2042 ms | 1952 ms | 1174 ms | Speech synthesis was the bottleneck      |

The synthesis delay was traced to the agent waiting for a whole sentence to be generated before playing any of it. It now streams during the filler. Fixed-line responses were additionally pre-recorded, removing a 1.5–2.5 s synthesis call from every policy turn.

### 9.2 Language model selection

The previous model was withdrawn by the provider mid-project (HTTP 404). Candidates were benchmarked on the project's own prompts before choosing a replacement.

| Model               | First spoken word   | Intent classification           | Outcome             |
|---------------------|---------------------|---------------------------------|---------------------|
| qwen/qwen3.8-27b    | 140–212 ms          | 195 ms, valid JSON, 3/4 correct | Selected            |
| openai/gpt-oss-120b | 500–890 ms          | not tested                      | Rejected — slower   |
| openai/gpt-oss-20b  | 216–715 ms, erratic | empty output, up to 8.7 s       | Rejected — unusable |

### 9.3 Other checks

| Check                                                      | Result                                        | Status |
|------------------------------------------------------------|-----------------------------------------------|--------|
| Configured models verified against the provider at startup | Reports missing models in logs and at /health | PASS   |
| System prompt size (previously 12–14k characters)          | Under 5,000 characters                        | PASS   |
| Fixed prompt prefix identical across turns (caching)       | Byte-identical                                | PASS   |
| Cached audio matches its recorded text                     | 125 clips verified against manifest           | PASS   |
| Suite repeatability                                        | 4 consecutive runs, identical results         | PASS   |

## 10. Manual live-call test

One end-to-end call through the browser client against the running server with live Deepgram, Groq and Rime.

| ID      | Step                                               | Observed result                                                      | Status |
|---------|----------------------------------------------------|----------------------------------------------------------------------|--------|
| TC-M-01 | Start the server                                   | Model availability check passed and was logged                       | PASS   |
| TC-M-02 | Connect and listen to the greeting                 | Cached greeting played; voice-activity model loaded                  | PASS   |
| TC-M-03 | "Yes. Now it's a good time."                       | Identity confirmed; reminder delivered                               | PASS   |
| TC-M-04 | "Can you tell me about the payment?"               | Explained; stated it had no amount on file rather than inventing one | PASS   |
| TC-M-05 | Interrupt NORA mid-sentence (three times)          | Speech stopped within 0.7 s each time                                | PASS   |
| TC-M-06 | "Fuck you."                                        | Recognised in 1 ms; one calm de-escalation offer                     | PASS   |
| TC-M-07 | "I just don't want to talk to you."                | One alternative offered                                              | PASS   |
| TC-M-08 | "As I said, I don't want to talk."                 | Call closed with a goodbye; disposition DECLINED                     | PASS   |
| TC-M-09 | Partial words while speaking ("Can you", "I just") | Discarded as fragments, not answered                                 | PASS   |

## 11. Defects found during testing and their fixes

| ID   | Defect                                                                                             | Found by  | Fix                                                                           | Retest |
|------|----------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------|--------|
| D-01 | Reply model withdrawn by the provider (HTTP 404) — also affected intent classification and warm-up | Live call | Switched to a benchmarked model; names made configurable; startup check added | PASS   |

| ID   | Defect                                                                                            | Found by                       | Fix                                                                                  | Retest |
|------|---------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------|--------|
| D-02 | Cached greeting spoke a different customer's name than the code intended                          | Review of the prerecord script | Greeting text centralised; manifest check added so mismatched clips are never played | PASS   |
| D-03 | Two-word confirmations such as "yes speaking" were discarded as fragments                         | Call simulation (TC-I-01)      | Short yes/no and control phrases now recognised as real turns                        | PASS   |
| D-04 | A word-boundary rule contained invisible control characters, so "just end the call" never matched | Policy test (TC-E-13)          | Characters replaced; whole repository scanned for others                             | PASS   |
| D-05 | Call could hang up silently if the model, speech synthesis or speech recognition failed           | Code review of all exit paths  | Every exit now speaks a goodbye, with a cached clip as the final fallback            | PASS   |
| D-06 | Fallback exit path passed an argument the model function does not accept                          | Code review                    | Signature corrected                                                                  | PASS   |
| D-07 | Voice-activity model could be called before loading, crashing the call                            | Code review                    | Lazy load with a safe fallback                                                       | PASS   |
| D-08 | Speech chunks ending in a comma produced an audible stumble (",.")                                | Live call                      | Trailing pause marks converted to sentence ends                                      | PASS   |
| D-09 | NORA offered to transfer the caller, which the system cannot do                                   | Live call                      | Prompt rule added: offer a callback, never a transfer                                | PASS   |
| D-10 | A "yes" to "Is this a good time?" wrongly confirmed identity                                      | Flaky simulation run           | Identity confirmation now requires an identity question and an explicit confirmation | PASS   |

## 12. Not covered

- Live telephone calls through Twilio or RingCX: the bridges were deliberately left unchanged, and only browser calls were tested end to end.
- The web front end: a known TypeScript build error from earlier interface work is outstanding and was out of scope.
- Load and concurrency: all tests use a single call at a time.
- Speech recognition accuracy across accents and noisy lines: only the one live call was observed.
- Live human transfer: by design, escalation records a callback request instead.

## Appendix A — Automated test inventory

Every automated test function, the number of data variations it runs, and its recorded result.

| Suite             | Test function                                                   | Cases | Result |
|-------------------|-----------------------------------------------------------------|-------|--------|
| intent_heuristics | test_bare_no_is_not_a_refusal                                   | 1     | PASS   |
| intent_heuristics | test_classify_intent_fast_path_precedence                       | 3     | PASS   |
| intent_heuristics | test_compliance_intents_exit_the_call                           | 1     | PASS   |
| intent_heuristics | test_goodbye_is_not_a_refusal                                   | 1     | PASS   |
| intent_heuristics | test_heuristic_intent                                           | 40    | PASS   |
| intent_heuristics | test_long_sentence_mentioning_bye_is_not_goodbye                | 1     | PASS   |
| intent_heuristics | test_numeric_answer_is_unclear_not_refusal                      | 1     | PASS   |
| policy            | test_abuse_deescalates_once_then_ends                           | 1     | PASS   |
| policy            | test_abuse_then_end_request_closes                              | 1     | PASS   |
| policy            | test_ai_question_is_answered_honestly                           | 1     | PASS   |
| policy            | test_all_templates_format                                       | 1     | PASS   |
| policy            | test_already_paid_closes                                        | 1     | PASS   |
| policy            | test_any_reply_ends_hold                                        | 1     | PASS   |
| policy            | test_anytime_closes_with_business_hours_promise                 | 1     | PASS   |
| policy            | test_busy_gets_one_offer_then_close                             | 1     | PASS   |
| policy            | test_busy_with_time_schedules_callback                          | 1     | PASS   |
| policy            | test_classify_telephony_prompt                                  | 4     | PASS   |
| policy            | test_compliance_stops_end_without_llm                           | 6     | PASS   |
| policy            | test_confirmation_after_greeting_verifies_identity              | 1     | PASS   |
| policy            | test_declining_callback_after_ai_disclosure_continues_the_call  | 1     | PASS   |
| policy            | test_distress_ends_with_crisis_line                             | 1     | PASS   |
| policy            | test_do_not_call_is_recorded_for_suppression                    | 1     | PASS   |
| policy            | test_escalation_declined                                        | 1     | PASS   |
| policy            | test_escalation_with_time_in_same_sentence_confirms_immediately | 1     | PASS   |
| policy            | test_escalations_offer_callback                                 | 5     | PASS   |
| policy            | test_every_call_ending_line_says_goodbye                        | 19    | PASS   |
| policy            | test_extract_callback_time                                      | 6     | PASS   |
| policy            | test_extract_callback_time_edge_cases                           | 8     | PASS   |
| policy            | test_good_time_question_is_not_identity_confirmation            | 1     | PASS   |
| policy            | test_goodbye_ends                                               | 1     | PASS   |
| policy            | test_hold_checkin_and_timeout                                   | 1     | PASS   |
| policy            | test_hold_keeps_the_callback_question_open                      | 1     | PASS   |
| policy            | test_loop_breaker_after_same_question_twice                     | 1     | PASS   |
| policy            | test_new_outcome_while_waiting_for_time_wins                    | 1     | PASS   |
| policy            | test_new_question_while_pending_callback_goes_to_llm            | 1     | PASS   |
| policy            | test_normal_turn_goes_to_llm                                    | 1     | PASS   |
| policy            | test_not_interested_then_no_closes_softly                       | 1     | PASS   |
| policy            | test_not_interested_then_stop_is_do_not_call                    | 1     | PASS   |
| policy            | test_okay_alone_does_not_confirm_identity                       | 1     | PASS   |
| policy            | test_partial_opt_out_records_preference                         | 1     | PASS   |
| policy            | test_pay_now_never_takes_payment_and_caps_callback_asks         | 1     | PASS   |
| policy            | test_reminder_marked_delivered_after_remind_turn                | 1     | PASS   |
| policy            | test_reminder_not_marked_delivered_without_reminder_content     | 1     | PASS   |
| policy            | test_repeat_keeps_the_callback_question_open                    | 1     | PASS   |

| Suite         | Test function                                              | Cases | Result |
|---------------|------------------------------------------------------------|-------|--------|
| policy        | test_repeat_replays_last_reply_then_offers_callback        | 1     | PASS   |
| policy        | test_reschedule_asks_for_time_then_confirms                | 1     | PASS   |
| policy        | test_slow_down_lowers_speed_and_repeats                    | 1     | PASS   |
| policy        | test_still_busy_without_time_closes                        | 1     | PASS   |
| policy        | test_third_party_gets_callback_without_disclosure          | 1     | PASS   |
| policy        | test_third_party_who_does_not_know_gets_polite_close       | 1     | PASS   |
| policy        | test_yes_with_screening_question_still_confirms_identity   | 1     | PASS   |
| prompt        | test_account_details_are_gated_on_identity                 | 1     | PASS   |
| prompt        | test_email_confirmation_instruction                        | 1     | PASS   |
| prompt        | test_fixed_prefix_is_identical_across_turns                | 1     | PASS   |
| prompt        | test_legacy_callers_without_stage_still_work               | 1     | PASS   |
| prompt        | test_no_duplicate_or_legacy_sections                       | 1     | PASS   |
| prompt        | test_prompt_builds_for_every_intent_and_stage              | 240   | PASS   |
| prompt        | test_prompt_is_much_smaller_than_before                    | 1     | PASS   |
| prompt        | test_prompt_requires_honest_ai_disclosure                  | 1     | PASS   |
| turn_flow     | test_intents_used_by_session_modes_exist                   | 1     | PASS   |
| turn_flow     | test_no_speculation_for_policy_handled_turns               | 6     | PASS   |
| turn_flow     | test_session_stage_progression_and_latency_marks           | 1     | PASS   |
| turn_flow     | test_speculation_matches_classifier_or_abstains            | 9     | PASS   |
| llm_failures  | test_first_failure_asks_caller_to_repeat                   | 1     | PASS   |
| llm_failures  | test_second_failure_stays_quiet_for_main_to_end_call       | 1     | PASS   |
| llm_failures  | test_wrap_up_failure_still_says_goodbye                    | 1     | PASS   |
| ws_simulation | test_busy_then_callback_time                               | 1     | PASS   |
| ws_simulation | test_confirm_then_already_paid                             | 1     | PASS   |
| ws_simulation | test_do_not_call_ends_immediately_and_records              | 1     | PASS   |
| ws_simulation | test_every_greeting_asks_who_is_speaking                   | 1     | PASS   |
| ws_simulation | test_goodbye_falls_back_to_cached_audio_when_tts_is_silent | 1     | PASS   |
| ws_simulation | test_repeated_llm_failure_ends_with_goodbye                | 1     | PASS   |
| ws_simulation | test_short_control_phrase_is_not_dropped_and_hold_resumes  | 1     | PASS   |
| ws_simulation | test_voicemail_leaves_message                              | 1     | PASS   |

Total: 409 automated cases, 409 passed, 0 failed.